Baby Universes from Thermal Pure States in the SYK model

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QuRIOS Seminar
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With Alejandro Vilar Lopez and Martin Sasieta (2512.00149) and Valerie Bettaque (coming soon)

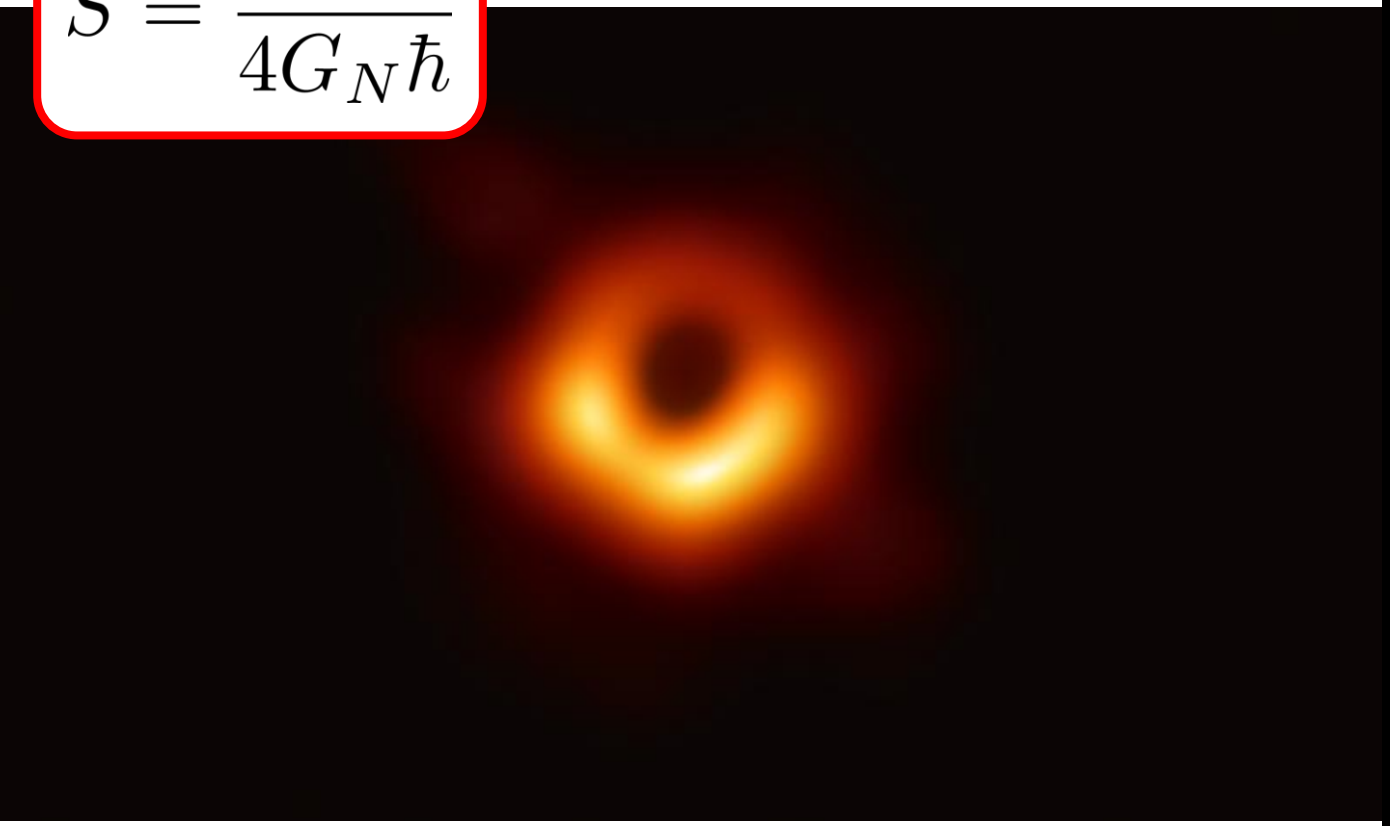
Alchemical ouroboros (Wikipedia)

Quantum gravity and cosmology

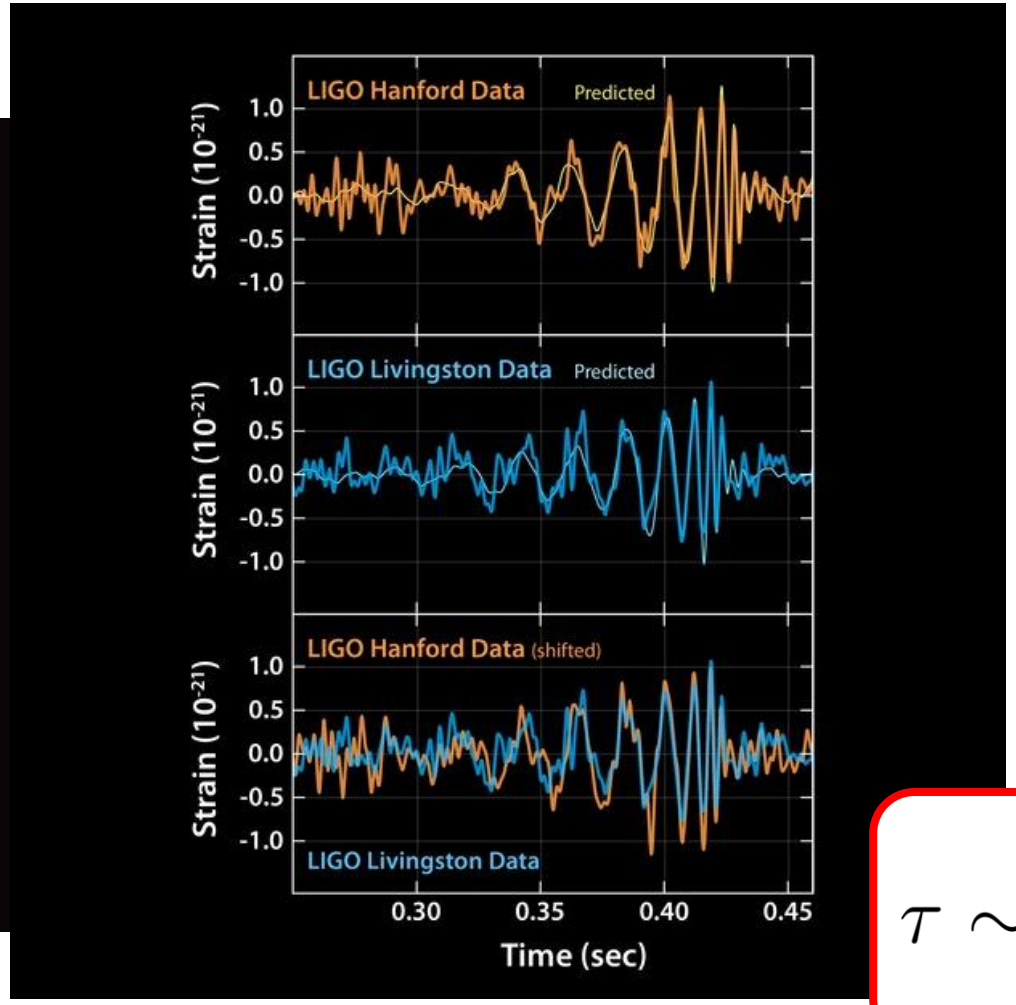
Quantum gravity is mysterious!

$$S = \frac{\text{Area}}{4G_N\hbar}$$

[Bekenstein, Hawking]



[Supermassive black hole in M87 – EHT, 2019]

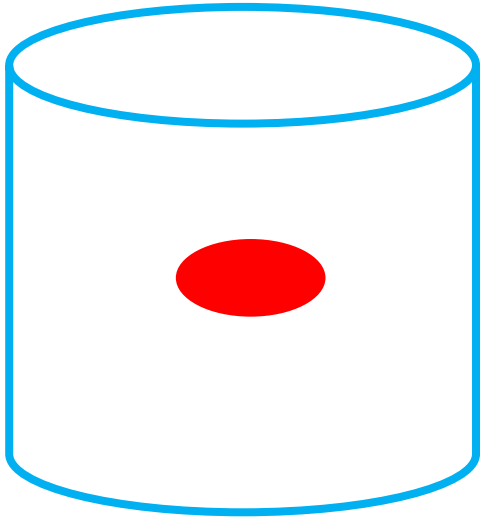


$$\tau \sim \frac{\hbar}{T}$$

AdS/CFT and "it from qubit"

For a "universe in a bottle" (Anti de Sitter space), the AdS/CFT correspondence gives a framework for quantum gravity: [Maldacena, ...]

- Wonderful theoretical lab to study black holes, strong coupling dynamics, hydrodynamics, ...
- Gravitational physics is intimately linked with quantum information: entanglement and geometry, complexity and black holes, ...
- By simulating the boundary system in a lab, we can hope to experimentally study holography!



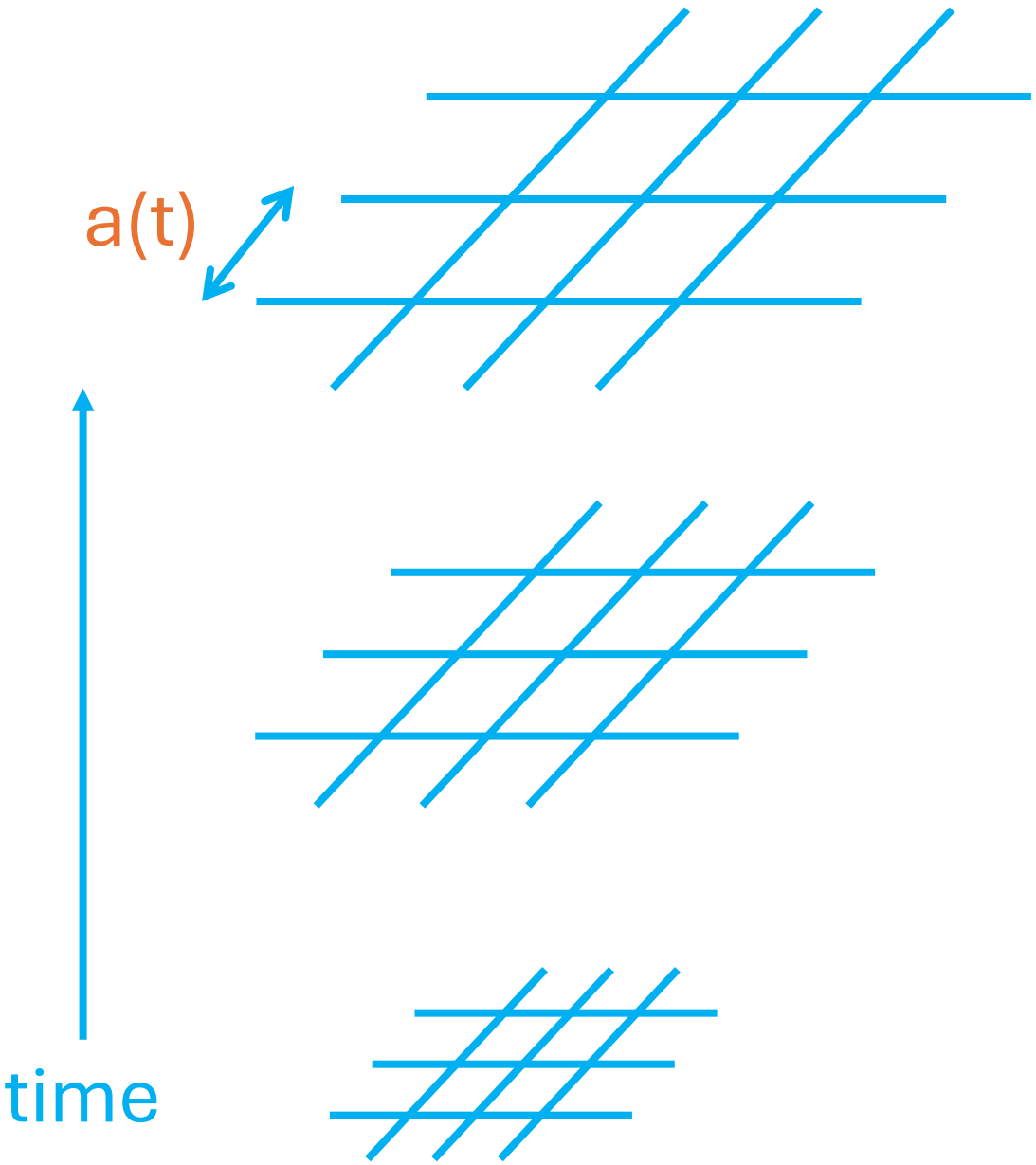
Stuff in AdS ←→
States in dual CFT

Cosmology

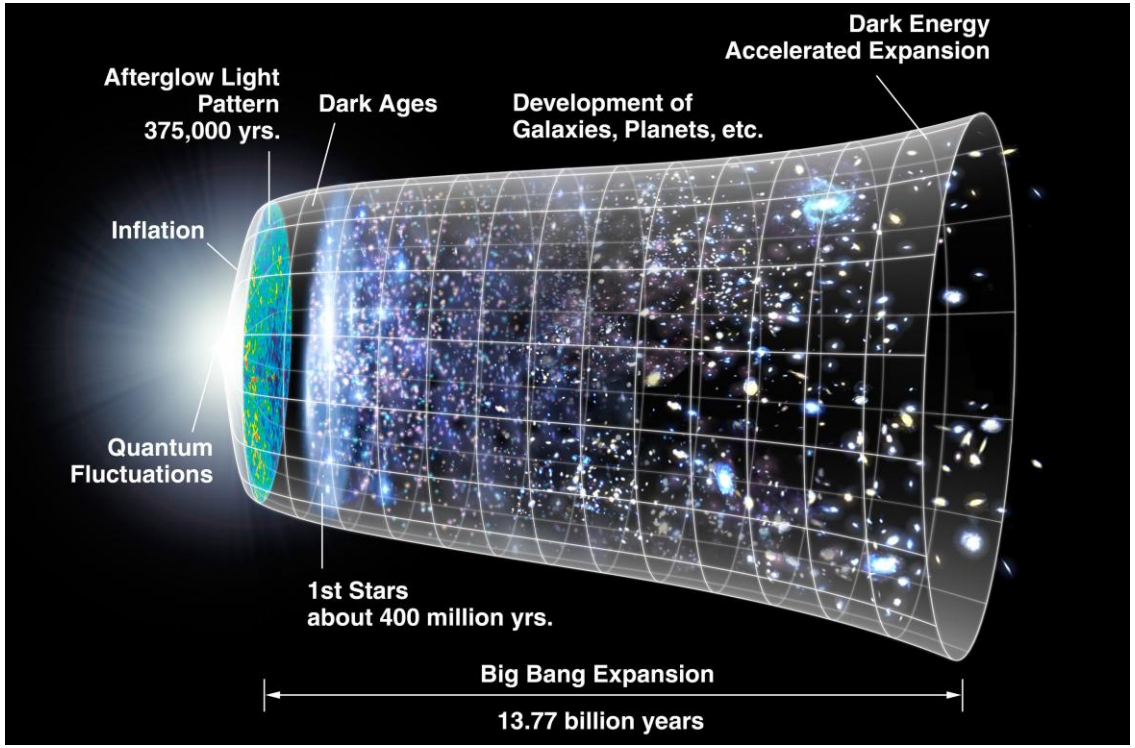
Our universe is:

- Full of stuff
- Nearly flat
- Homogenous on large scales
- Expanding: a(t)

There is no obvious boundary!



Standard picture



Cosmological constant

Cold dark matter

Supernovae + CMB + … → ΛCDM

- The expansion is accelerating
- Dark energy takes over $\Lambda>0$
- Universe approaches de Sitter (maybe a future boundary?)

Alternatives?

Is $\Lambda > 0$ dS-like the only interesting option?

- Data: A good model, but there are wrinkles …
- Conceptually: No! We can have Big Bang cosmologies with $\Lambda < 0$

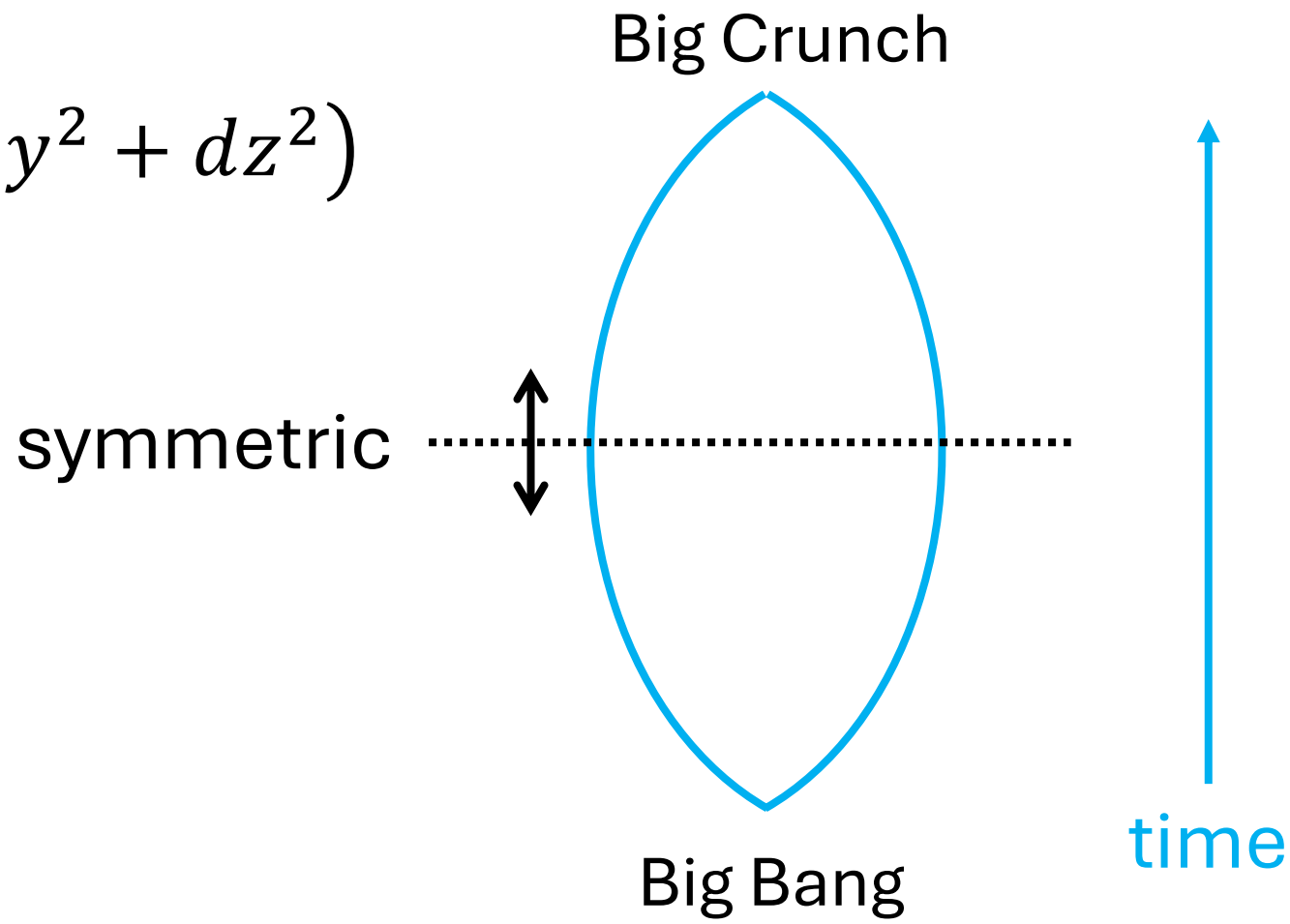
From the bulk perspective in AdS/CFT, we have some gravitational effective field theory with $\Lambda < 0$ and these have cosmological solutions

Example

Flat + radiation + $\Lambda < 0$

$$ds^2 = -dt^2 + a(t)^2\left(dx^2 + dy^2 + dz^2\right)$$

$$a(t)^2 = \cos t/t_*$$



No AdS boundaries!

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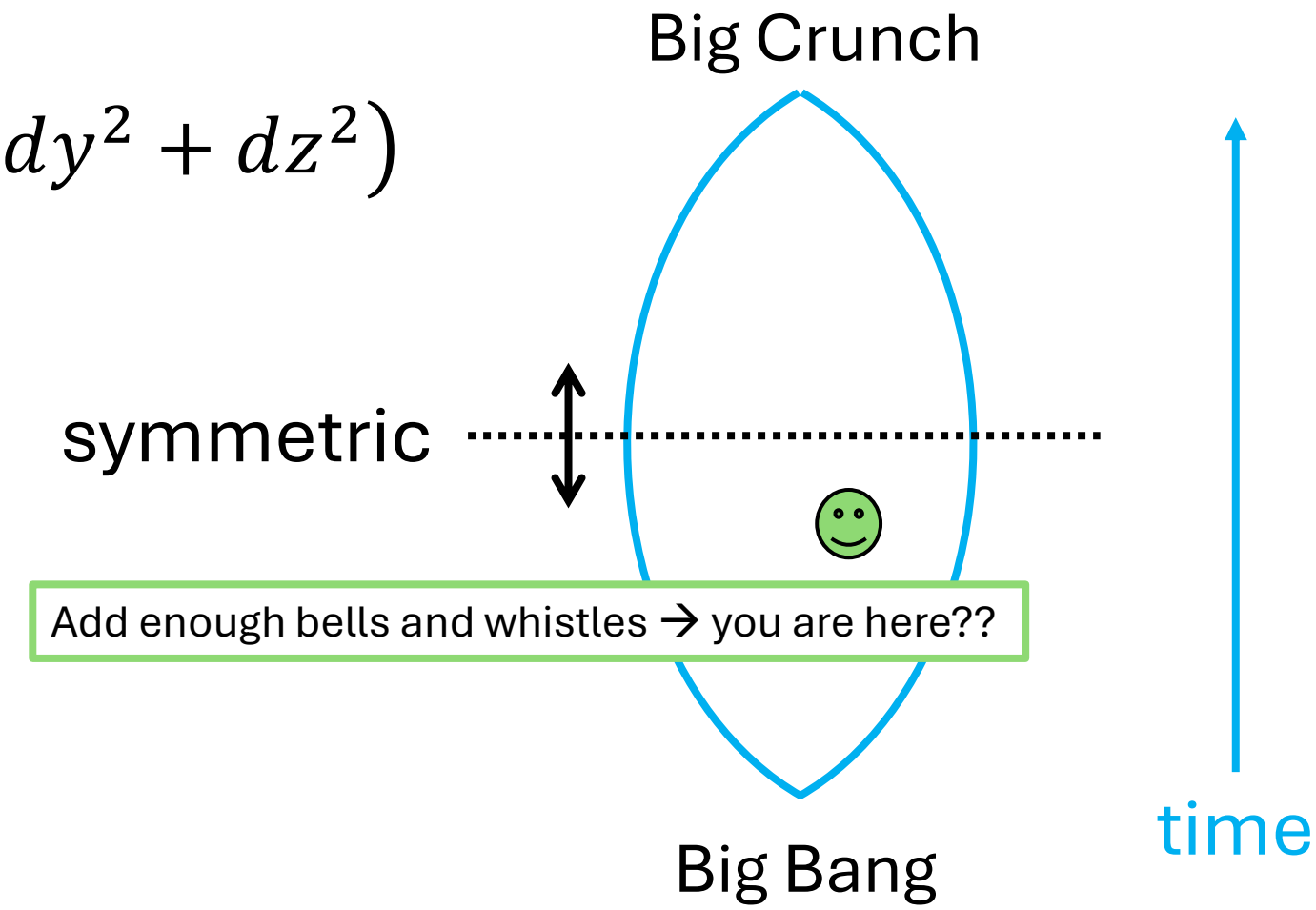
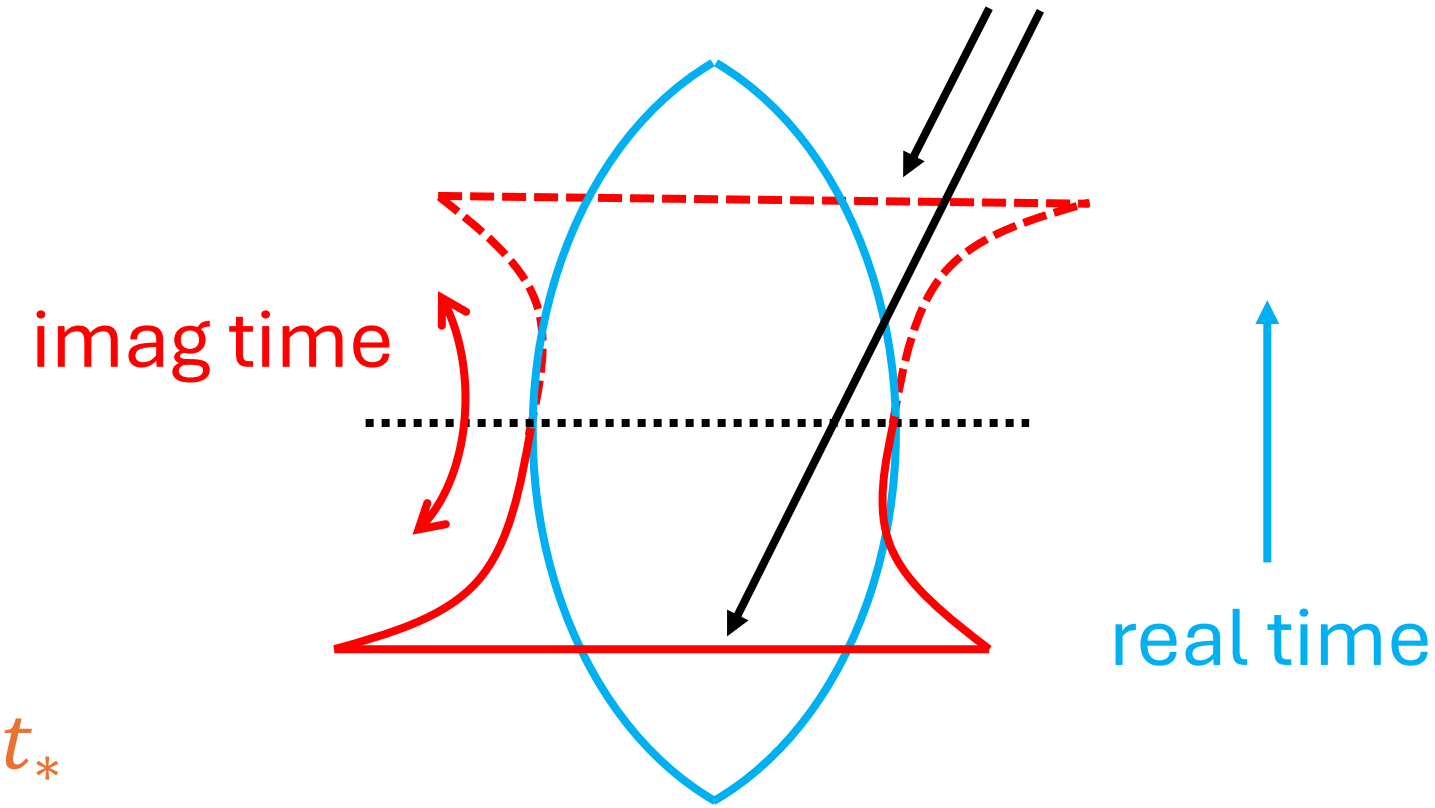


No AdS boundaries!

[more realistic models: Antonini-Waddell-Simidzija-Van Raamsdonk-S, ...]

Meaning?

Euclidean wormhole connecting two distinct AdS boundaries



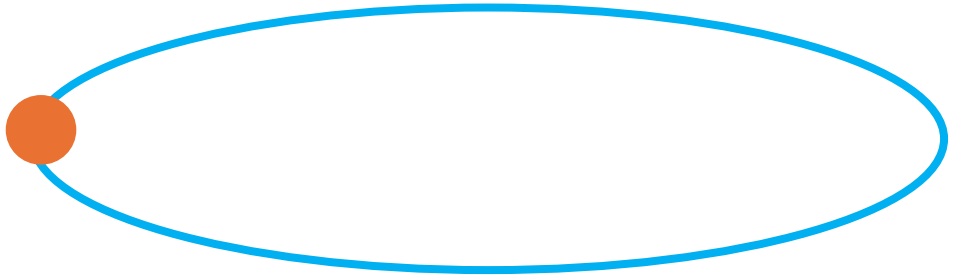
$$a(t)^2 = \cos t/t_*$$

$$t \to i\tau: a(\tau)^2 = \cosh \tau/t_*$$

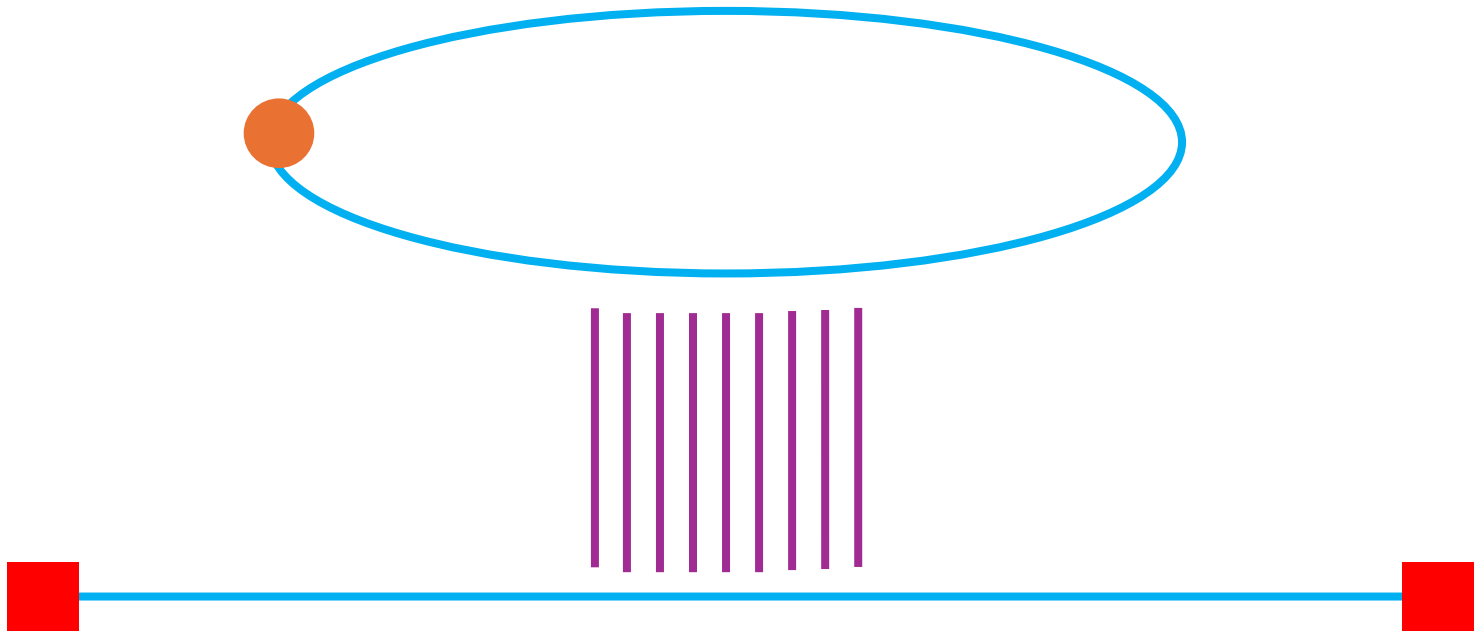
[Maldacena-Maoz, McInnes, Antonini-Waddle-Simidzija-Van Raamsdon-S, ...]

Baby universes from SYK

The baby universe



The space of the closed universe consists of a mostly empty circle with a heavy clump of matter



There are also quantum fields in the closed universe which can be entangled with fields in a separate spacetime region with asymptotic AdS boundaries → this gives a state in the Hilbert space associated with the boundaries

[a version of AS^2: Antonini-Sasieta-S, ...]

Microscopic model

- We start with the Sachdev-Ye-Kitaev (SYK) model on N fermions:

$$\{\psi_i, \psi_j\} = 2\delta_{ij}$$

$$H_{\mathrm{SYK}} = \mathrm{i}^{\frac{q}{2}} \sum_{i_1 < \ldots < i_q} J_{i_1 \ldots i_q} \psi_{i_1} \ldots \psi_{i_q}$$

$$\overline{J_{i_1 \ldots i_q} J_{i'_1 \ldots i'_q}} = \frac{\mathcal{J}^2 N}{2q^2 \binom{N}{q}} \delta_{i_1, i'_1} \ldots \delta_{i_q, i'_q}$$

Microscopic model

- We also need two additional ingredients:
  - (i) A first order phase transition to effectively evaporate a black hole, thus disconnecting its interior which becomes the closed universe
  - (ii) A heavy operator to support the closed universe

For (i), we use the Maldacena-Qi Hamiltonian:

$$H = H_{\mathrm{SYK,L}} + H^*_{\mathrm{SYK,R}} - \frac{\mu N}{\binom{N}{p}} (-\mathrm{i})^{p^2} \sum_{i_1 < \ldots < i_p} \psi^{\mathsf{L}}_{i_1 \ldots i_p} \psi^{\mathsf{R}}_{i_1 \ldots i_p}$$

For (ii), we use a many-fermion operator:

$$\mathcal{O}_\Delta = \mathrm{i}^{\frac{r}{2}} \psi_{i_1} \ldots \psi_{i_r}$$

[Sasieta-S-Vilar Lopez]

Microscopic state

heavy operator acting on left side of maximally entangled LR state

$$|\mathcal{O}_\Delta\rangle \equiv \mathcal{O}^{\mathsf{L}}_\Delta |I\rangle$$

- Thermal pure state:

$$|\Psi^\Delta_\beta\rangle \propto e^{-\frac{\beta}{2}H} |\mathcal{O}_\Delta\rangle = \sum_n e^{-\frac{\beta}{2}E_n} c_n |E_n\rangle$$

thermal cooling with MQ Hamiltonian

erratic coefficients

energy eigenstates of MQ Hamiltonian

- This is the microscopic state of the coupled SYK dots; it varies erratically with the SYK couplings and the choice of operator and its statistical properties encode the closed universe physics

[Sasieta-S-Vilar Lopez]

What do we compute?

$$\mathcal{N}_\beta^\Delta = \langle \mathcal{O}_\Delta | \, e^{-\beta H} \, | \mathcal{O}_\Delta \rangle = \sum_n e^{-\beta E_n} |c_n|^2$$

ensemble average $$\overline{\mathcal{N}_\beta^\Delta} \sim e^{-I_\beta[\Sigma_\mathrm{d}]} Z_\mathrm{d} + e^{-I_\beta[\Sigma_\mathrm{h}]} Z_\mathrm{h}$$

We can develop a path integral expression for this quantity either directly in the SYK language or using a corresponding bulk theory

[Sasieta-S-Vilar Lopez]

Semi-classical bulk picture

JT gravity coupled to matter, action: $I = I_{\mathrm{JT}} + I_{\mathrm{matter}}$

$$I_{\mathrm{JT}} = -S_0\,\chi[\Sigma] - \frac{1}{2}\int_{\Sigma}\Phi(R+2) - \int_{\partial\Sigma}\Phi_b(K-1)$$

Field content:

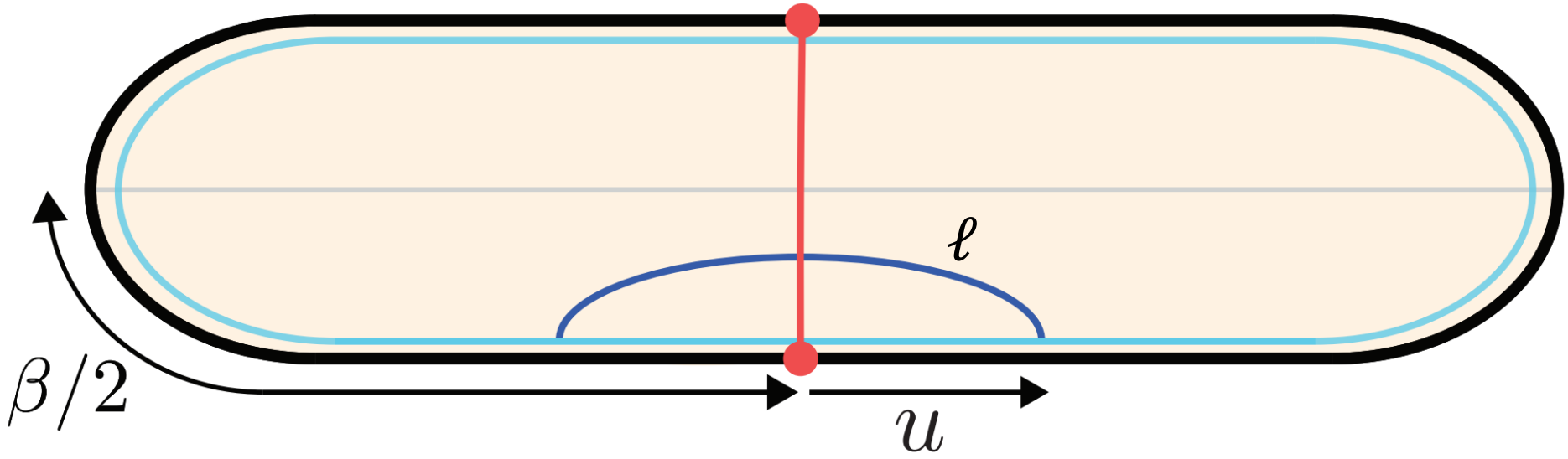
- The metric is fixed to be constant curvature,
- The dilaton (Phi) can have a non-trivial profile
- The matter can be taken to be O(N) massive fermion fields

[Kitaev, Sachdev, Maldacena-Stanford, Rosenhaus-Polchinski,...]

Two saddles

$$\overline{\mathcal{N}_\beta^\Delta} \sim e^{-I_\beta[\Sigma_\mathrm{d}]} Z_\mathrm{d} + e^{-I_\beta[\Sigma_\mathrm{h}]} Z_\mathrm{h}$$

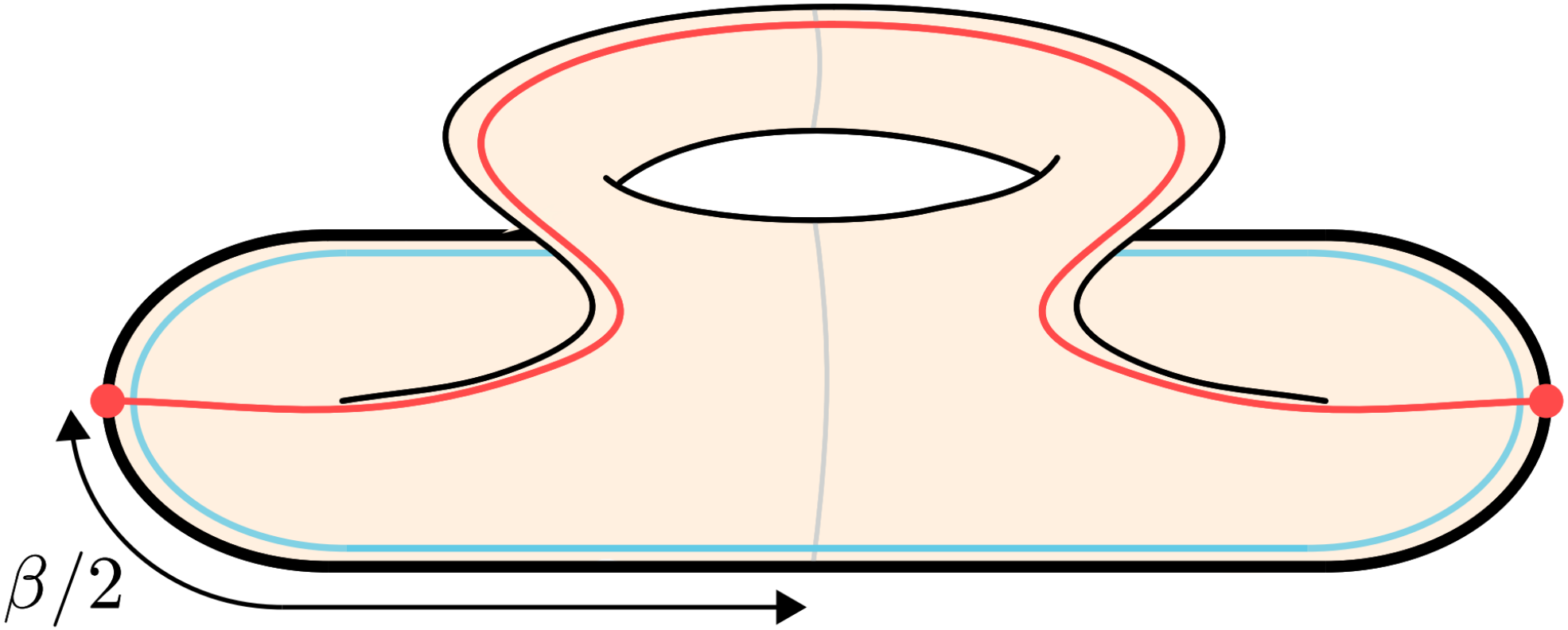
- Black hole saddle (Euclidean)



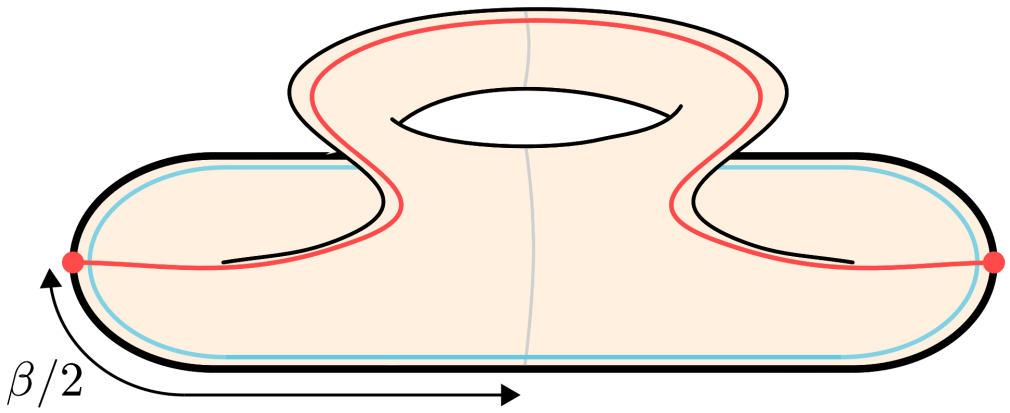
$$I = \frac{N}{2}\int \mathrm{d}u \left(\frac{1}{2}\dot{\ell}^2 + V(\ell)\right) - S_0 \qquad V(\ell) = 2e^{-\ell} - \eta e^{-\Delta_H \ell} + \frac{2\Delta}{N} e^{-\ell/2}$$

[Maldacena-Qi, Sasieta-S-Vilar Lopez]

- Baby universe saddle (Euclidean)

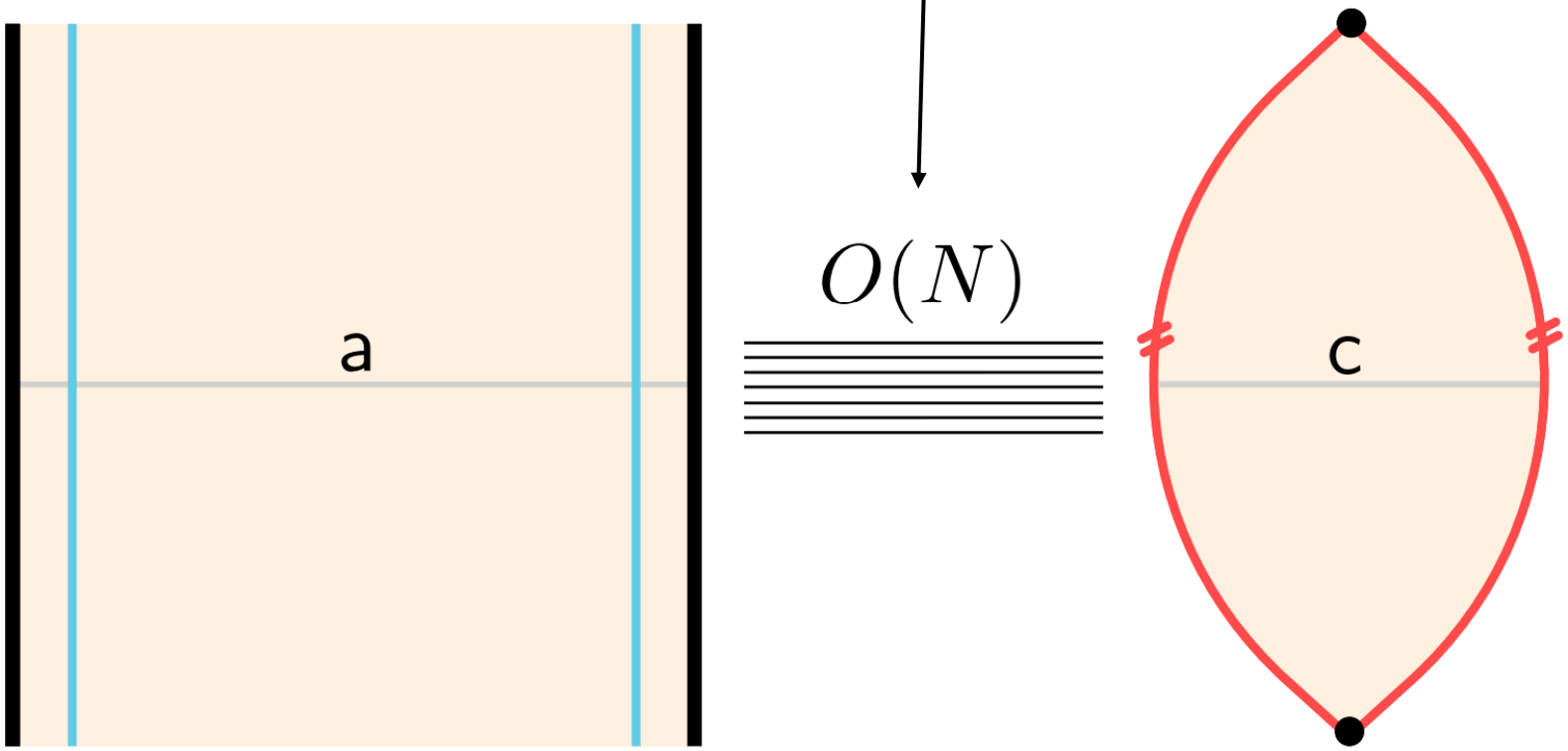


[Sasieta-S-Vilar Lopez]



one way to interpret the entanglement with the closed universe is in terms of the entropy of the ensemble-averaged state

Lorentzian picture:



Zero MQ temperature: an even simpler limit?

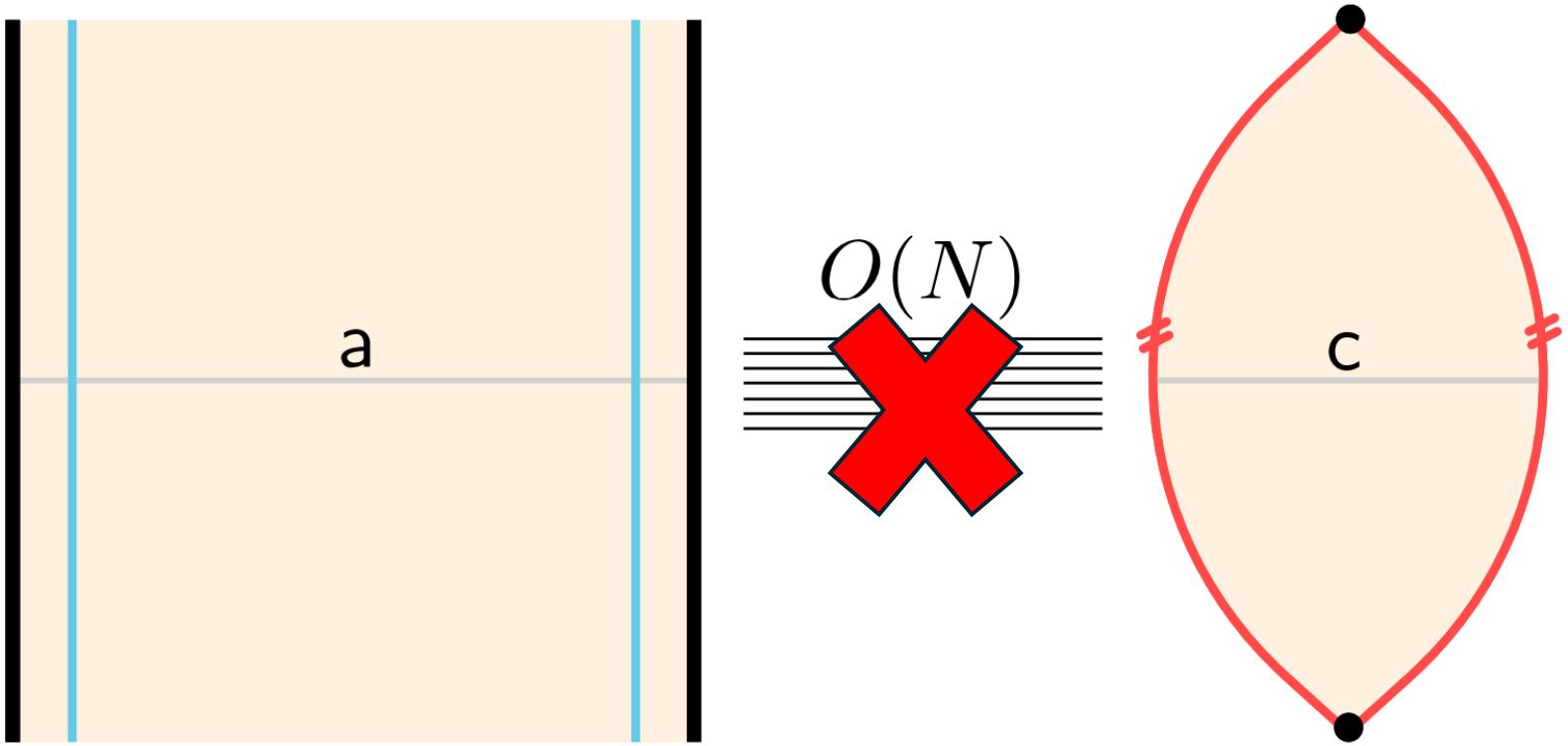
- If we take the beta parameter to infinity, then the microscopic state becomes proportional to the MQ ground state
- The MQ ground state is close to the thermofield double (TFD) state of the original SYK at a coupling dependent temperature

$$H = H_{\mathrm{SYK,L}} + H^*_{\mathrm{SYK,R}} - \frac{\mu N}{\binom{N}{p}} (-\mathrm{i})^{p^2} \sum_{i_1 < \ldots < i_p} \psi^{\mathrm{L}}_{i_1 \ldots i_p} \psi^{\mathrm{R}}_{i_1 \ldots i_p}$$

[Maldacena-Qi, Cottrell et al.]

- The SYK TFD is not erratic and isn't dual to a closed universe???

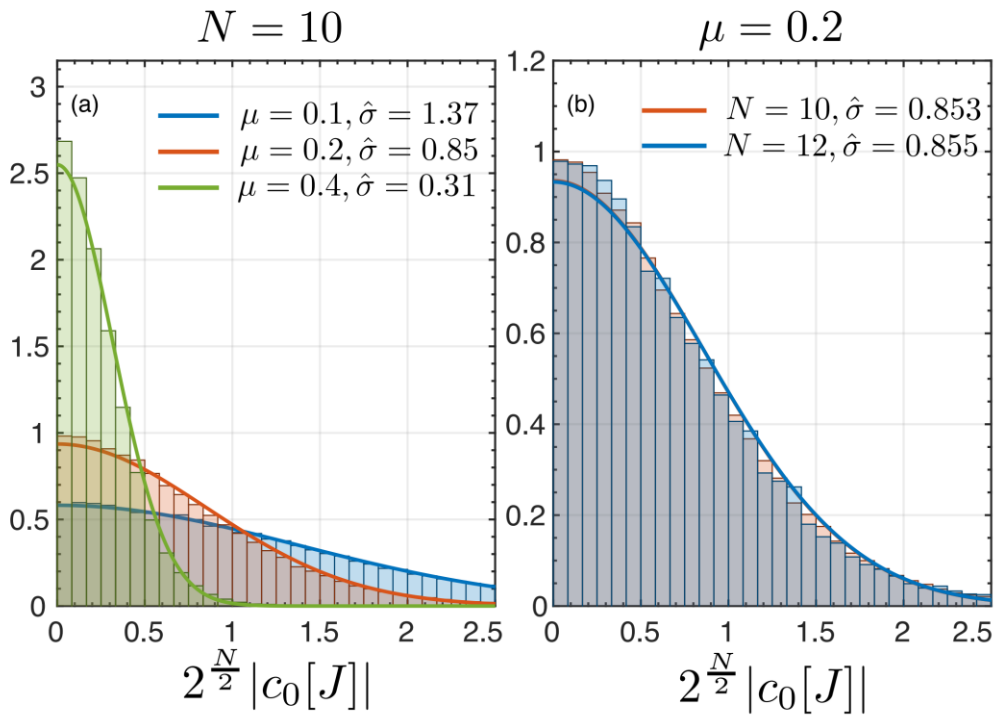
What's going on the bulk?



We're zeroing out the bulk entanglement with the closed universe, so the closed universe is floating "untethered" in the void

There is still something erratic …

- The overlap between the MQ ground state and the heavy operator state is actually erratic → the closed universe is hiding in the statistical properties of this matrix element!



At least at small N, the overlap behaves as a real gaussian random variable

[Sasieta-S-Vilar Lopez]

Hiding in what sense?

- This overlap is actually something simple in disguise, a thermal 1-point function: $\mathrm{tr}\left(e^{-\beta H_{syk}} O\right)$
- We can study the statistics of this object to see the emergence of the baby universe in its simplest form
- With Valerie Bettaque, we have studied this object in SYK in great detail (I will return to the bulk picture in a moment); this is also related to something called quantum magic

$$\mathcal{O}_{\Delta} = \mathrm{i}^{\frac{r}{2}} \psi_{i_1} \ldots \psi_{i_r}$$

Path integral approach

$$\mathrm{tr}\left(\mu(a)\, e^{-\beta H_{\mathrm{SYK}}}\right)^R = \mathrm{tr}\left[\mu^{\otimes_f R} \exp(-\beta H_{\mathrm{SYK}})^{\otimes_f R}\right]$$

$O \to \mu, \psi \to \chi$

$$= \mathrm{tr}\left[\mu^{(1)}(a)\, \mu^{(2)}(a) \cdots \mu^{(R)}(a) \exp\left(-\beta \sum_{r=1}^{R} H_{\mathrm{SYK}}^{(r)}\right)\right]$$

$$\simeq \mathrm{tr}\left[(-1)^{a_1 F_{R,1}} \cdots (-1)^{a_N F_{R,N}} \exp\left(-\beta \sum_{r=1}^{R} H_{\mathrm{SYK}}^{(r)}\right)\right]$$

$$(-1)^{F_{R,i}} := (2i)^{R/2}\, \chi_i^{(1)} \cdots \chi_i^{(R)}$$

[Bettaque-S]

Path integral approach

$$G_{rs}(\tau,\tau') := \frac{1}{N}\sum_{i=1}^{N}\chi_i^{(r)}(\tau)\,\chi_i^{(s)}(\tau'), \quad G_{rs}(\tau,\tau') = -G_{sr}(\tau',\tau),$$

multi-replica collective field

$$\overline{\mathrm{tr}(\mu(a)\,e^{-\beta H_{\mathrm{SYK}}})^R}$$

$$= \int [\mathcal{D}\Sigma]\,[\mathcal{D}G]\,\mathrm{pf}[I_R\otimes\mathcal{D}^- - \Sigma]^{N-W}\,\mathrm{pf}[I_R\otimes\mathcal{D}^+ - \Sigma]^{W}$$

$$\exp\left(-\frac{N}{2}\sum_{r,s=1}^{R}\int_0^\beta d\tau d\tau'\left(\Sigma_{rs}(\tau,\tau')\,G_{rs}(\tau,\tau') - \frac{J^2}{q}\,G_{rs}(\tau,\tau')^q\right)\right)$$

ensemble average washes out the dependence on which fermions were chosen for μ

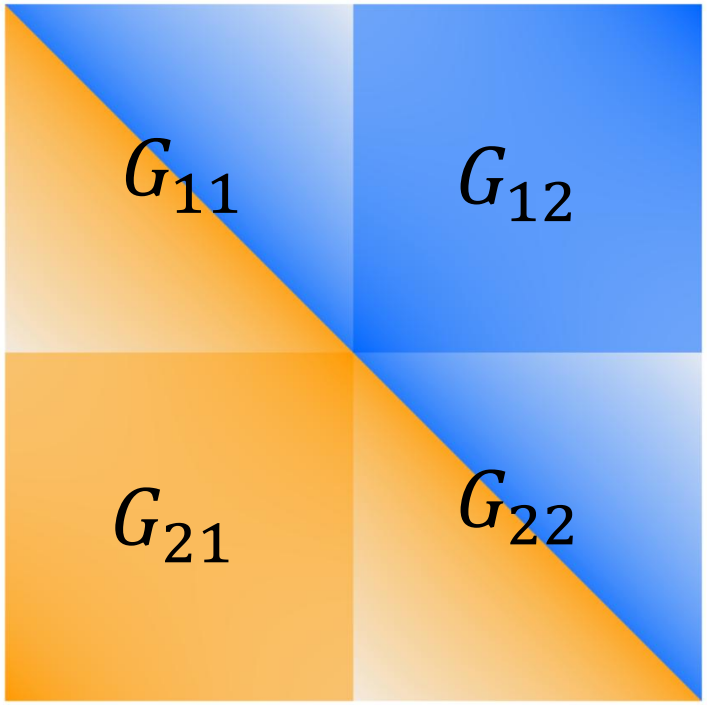
[Bettaque-S]

Path integral results

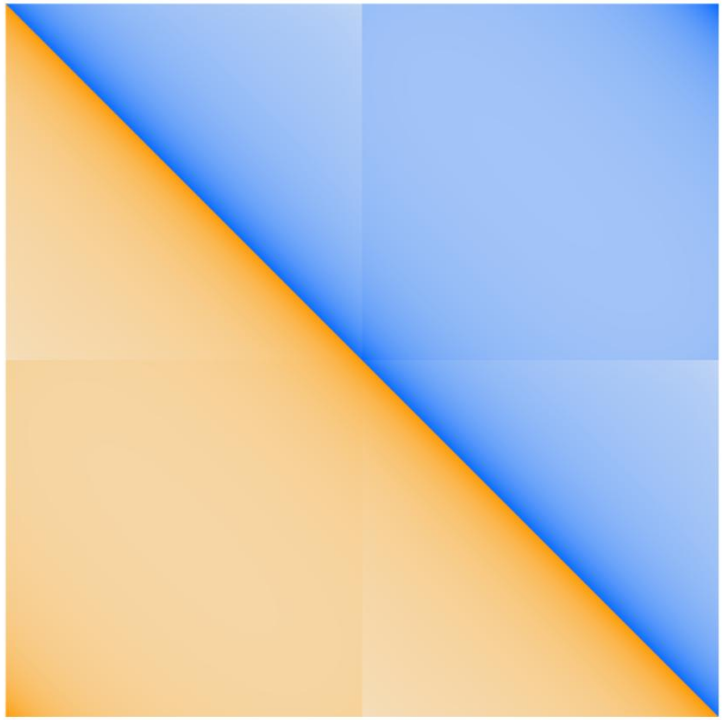
R=4

- Demanding numerics (replicas, broken time-translation, ...), but we find the 1-point functions are indeed real gaussian at large N

R=2



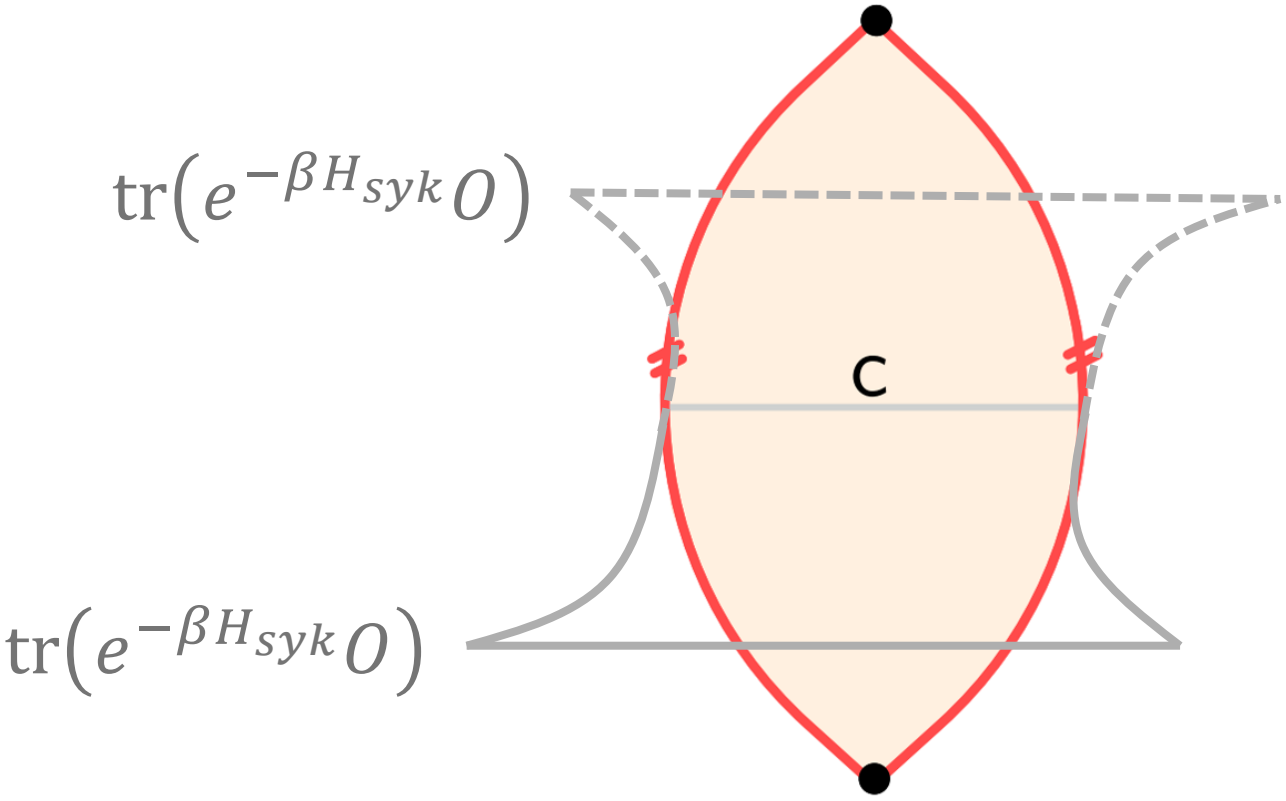
(b) $\beta = 5, w = 0.5$



(e) $\beta = 20, w = 0.5$

[Bettaque-S]

Where is the closed universe?



The inter-replica correlations, G_{21}, are correlations between the two sides of a Euclidean wormhole; analytic continuation gives the closed universe

[JT wormhole/cosmology solutions: Stanford, Usatyuk-Wang-Zhao]

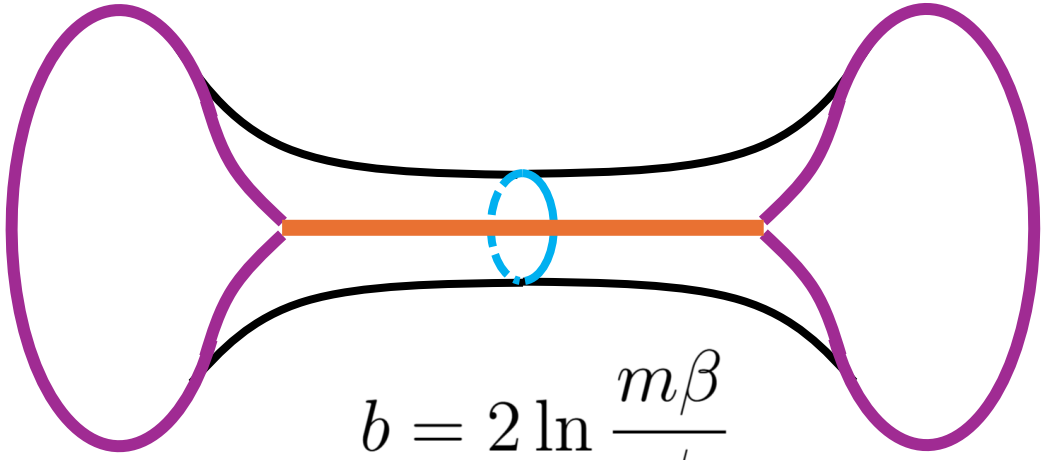
Wormhole calculation

- We model the SYK inter-replica correlation with a boundary-to-boundary correlator of a light probe fermion moving in a background geometry stabilized by the heavy particle [Bettaque-S]
- For simplicity, we use a geodesic approximation to estimate the bulk 2-point function of the probe fermion
- Background – double trumpet

$$ds^2 = d\rho^2 + b^2 \cosh^2 \rho \, d\sigma^2$$

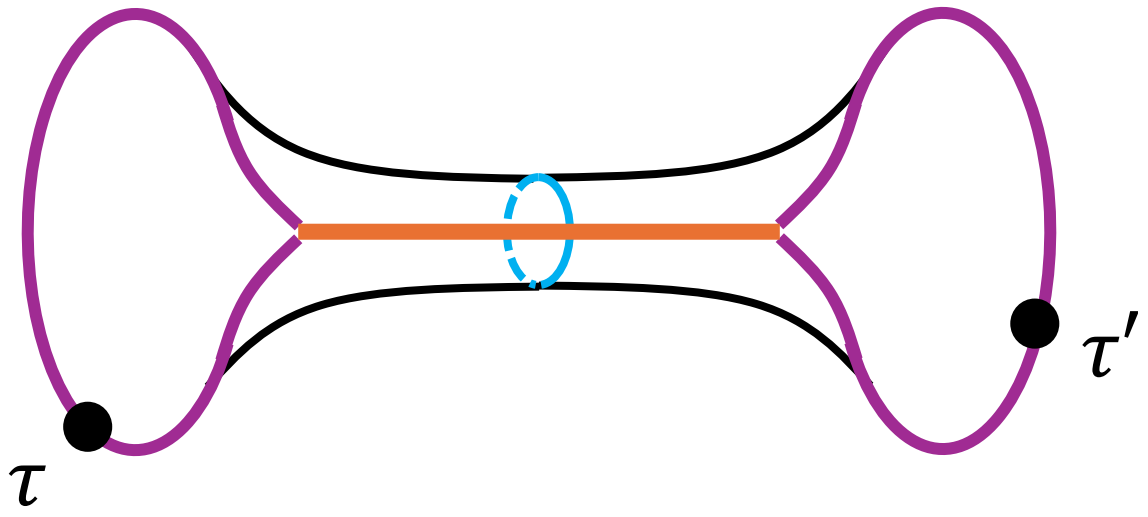
$$\Phi = \frac{\pi \phi_r}{\beta} \cosh \rho \cosh b\sigma.$$

[Stanford, Usatyuk-Wang-Zhao]



$$b = 2 \ln \frac{m\beta}{\pi \phi_r}$$

Wormhole calculation



$$e^{\rho_c} = \frac{2\beta}{\pi\epsilon}\frac{1}{\cosh b\sigma}.$$

$$\frac{\tau(\sigma)}{\beta} = \frac{\arctan e^{b\sigma} - \arctan e^{-b/2}}{\arctan e^{b/2} - \arctan e^{-b/2}}.$$

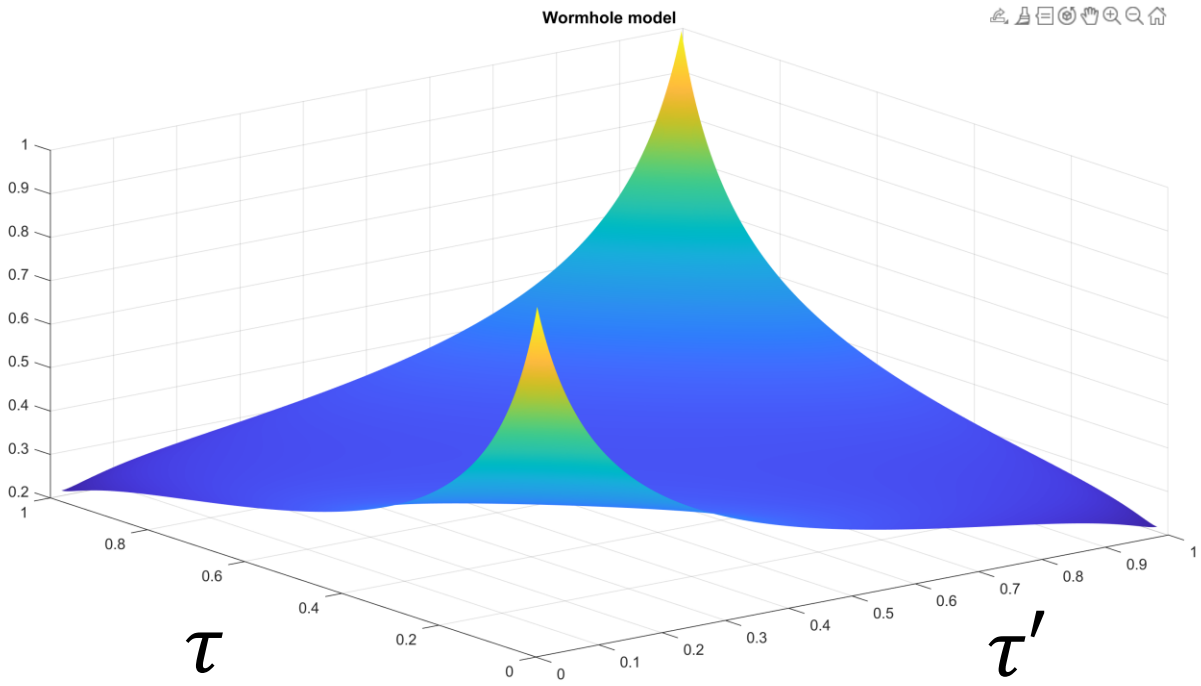
winding geodesics $d_n(\sigma,\sigma') = \rho_c(\sigma) + \rho_c(\sigma') + \log\cosh^2\frac{b(\sigma-\sigma'-n)}{2}.$

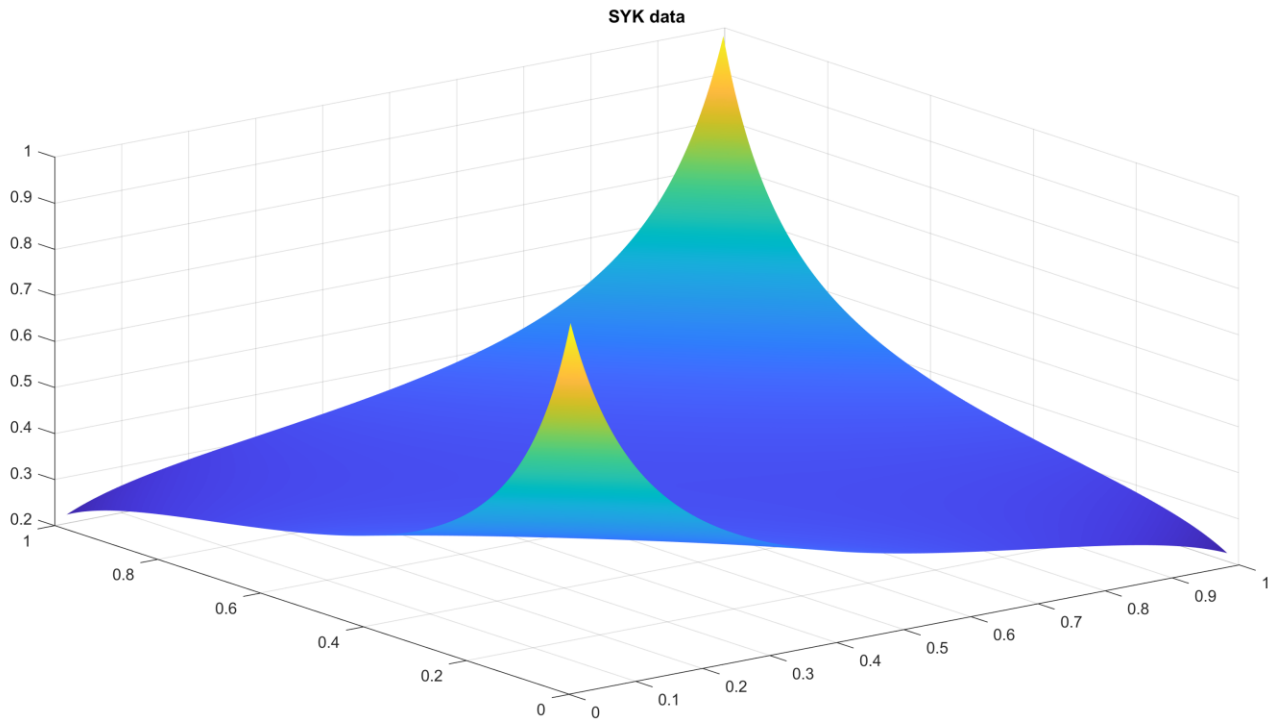
$$\mathcal{G}_P(\tau,\tau') = \mathcal{N}_P\sum_n e^{-\Delta d_n(\sigma(\tau),\sigma'(\tau'))}, \quad \mathcal{G}_{AP}(\tau,\tau') = \mathcal{N}_{AP}\sum_n(-1)^n e^{-\Delta d_n(\sigma(\tau),\sigma'(\tau'))},$$

[Bettaque-S]

Comparison of normalized correlations

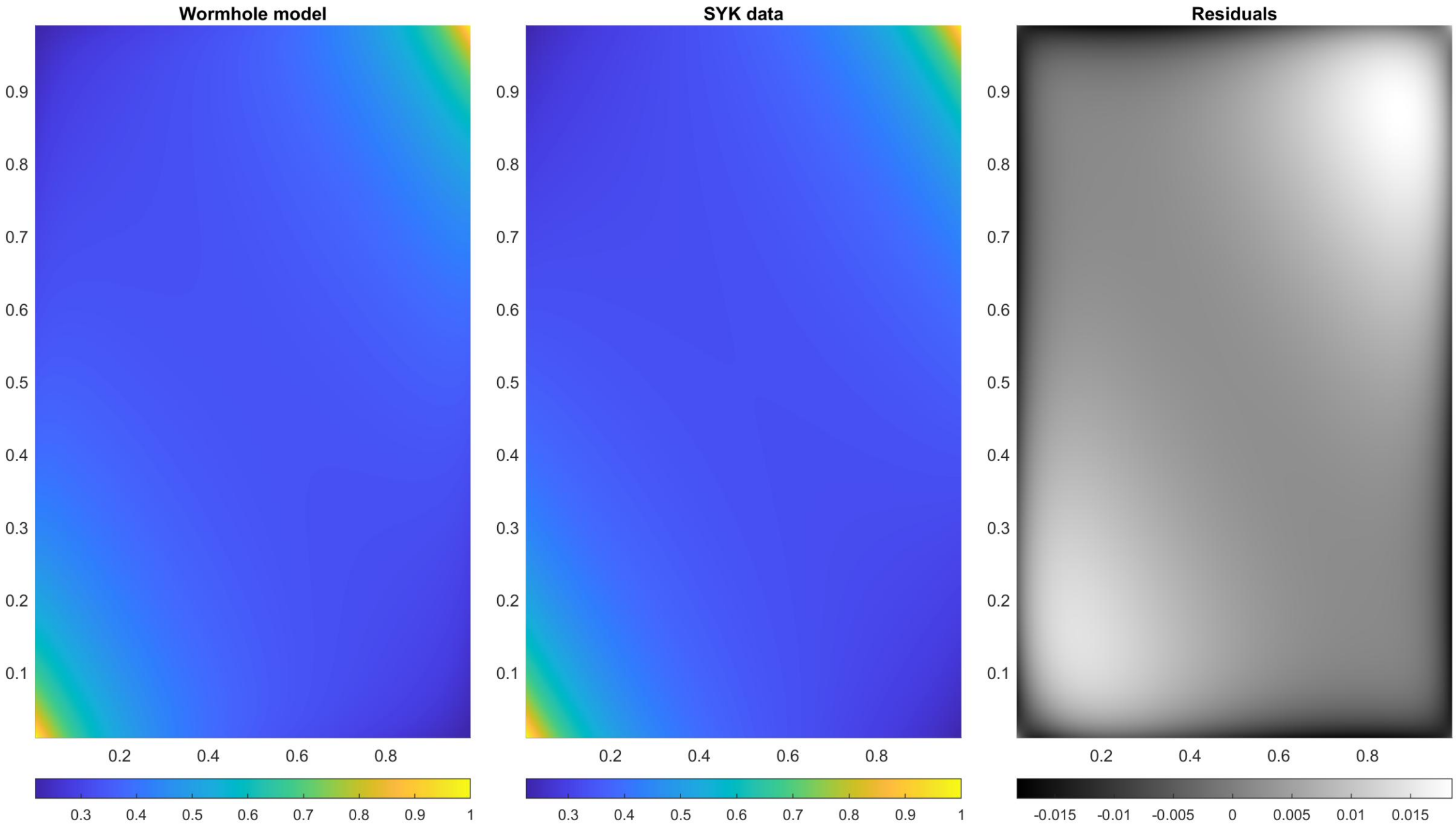
$$\mathcal{G}_f = (1-f)\mathcal{G}_{AP} + f\mathcal{G}_P.$$





Bulk model: $\mathrm{b} = 6.4, \mathrm{f} = .32$; SYK: $\mathrm{w} = .25, \beta\mathrm{J} = 20, \mathrm{L} = 1000$

[Bettaque-S]


Wormhole model
SYK data
Residuals
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.2
0.4
0.6
0.8
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
-0.015
-0.01
-0.005
0
0.005
0.01
0.015

So where are we?

- We have a match between averaged quantities in a microscopic theory and cosmological solutions in semi-classical gravity
- The erratic nature of the microscopic matrix elements and statistical averaging was crucial to reveal the closed universe
- This is a very active field at the moment, e.g. observers, debates about saddles, semiclassicality, ...

[averaging/algebras: Liu, Kudler Flam-Witten, Gesteau-Engelhardt, Gesteau, observers: Engelhardt-Gesteau-Harlow, Abdalla-Antonini-Iliesiu-Levine, Harlow-Usatyuk-Zhao, saddles: Vilar Lopez-Van Raamsdonk, averaging/semiclassicality: Antonini-...-S, large q SYK ...]

Looking forward

- We can dream, for example, about computing the bulk wavefunction at the time-symmetric slice to determine what observers in the universe would see if they looked at their "CMB"
- We can engage with data, and look for general lessons, e.g. in our setup the state of the universe is naturally defined at its midpoint, addresses cosmic coincidence, predicts eventual deceleration
- There is also more to understand about averaging, and how it interacts with more familiar forms of holography and observers
- Finally, we can explore these observables in quantum computers and look for quantum advantages in studying quantum cosmology

$$G_{21} \sim \overline{tr\left[exp\left(-\beta H_{syk}\right)O\chi(\tau)\right]tr\left[exp\left(-\beta H_{syk}\right)O\chi(\tau')\right]}$$